Pratyush Jha

Pre-doctoral researcher, Indian Institute of Science, Bangalore
+91 96543 50759 · pratyushjha254@gmail.com · pratyushjha@iisc.ac.in
Google Scholar · LinkedIn

Research interests. Statistical physics of generative models; using physical systems with exactly known or well-characterized statistics (spin models, turbulence) as controlled testbeds for what neural networks learn and where they fail; scientific machine learning.

Education

Indian Institute of Technology Roorkee Nov 2020 – Jul 2024
B.Tech. in Engineering Physics CGPA 9.117/10 · Department Rank 2

- **Thesis:** *Deep Learning Simulations for Phase Diagrams of Interacting Josephson Qubits* (Advisor: Prof. Peayush Choubey).
- **Relevant coursework:** Thermal & Statistical Physics, Stochastic Processes, Statistical Inference, Numerical Analysis & Computational Physics, Mathematical Physics, Quantum Physics, Quantum Computing, Elements of Condensed Matter Physics, Quantum Theory of Solids, Introduction to Machine Learning, Reinforcement Learning.

Publications & Preprints

[1] **P. Jha***, B. Maji*, R. Pandit. *How well can Diffusion Models learn Lagrangian-Tracer Statistics in Non-reciprocal Turbulence?* Preprint, arXiv:2608.27378 (Aug 2026). **Equal contribution.*

[2] **P. Jha**. *Coherence at the Financial Misinformation Detection Challenge Task: A Courtroom-Inspired Framework for Reference-Free Verification.* Workshop on Misinformation Detection, ICWSM 2026 (Jun 2026), DOI: 10.36190/2026.40. *Sole author; 3rd place in the shared task.*

[3] **P. Jha**. *Capturing the Topological Phase Transition and Thermodynamics of the 2D XY Model via Manifold-Aware Score-Based Generative Modeling.* Preprint, arXiv:2602.07548 (Feb 2026). *Sole author.*

Research Experience

Indian Institute of Science, Bangalore May 2026 – Present
Project Associate (Pre-doctoral Research) Advisor: Prof. Rahul Pandit

Generative diffusion models for non-reciprocal turbulence [arXiv:2608.27378]

- First application of generative diffusion models to non-reciprocal binary-fluid turbulence (2D Non-Reciprocal Cahn–Hilliard–Navier–Stokes), benchmarking synthetic Lagrangian-tracer trajectories against direct numerical simulations.
- Characterized Lagrangian multiscaling (velocity increments, their structure functions and flatnesses, multiscaling exponent ratios) for the first time in non-reciprocal hydrodynamics, alongside velocity and acceleration PDFs, curvature statistics, irreversibility and model transferability.
- Found that the accelerated DDIM sampler falters below 100 steps on some statistics, although it stays accurate down to 25 steps in 3D Navier–Stokes turbulence (Li *et al.*, 2026); an 800-step DDPM reconstructs the flow statistics accurately.

Independent Research Jan 2026 – Apr 2026
Sole-author projects

Manifold-aware score-based generative modeling of the 2D XY model [arXiv:2602.07548]

- Performed score-based diffusion directly on the torus (the 64×64 XY lattice, a 4096-dimensional torus); the learned score matches the theoretical Boltzmann score more precisely than a Euclidean-embedding diffusion model, which spends capacity learning the manifold constraint.
- To our knowledge, the first data-driven generative model (no Hamiltonian, no feature engineering) to reproduce the heat capacity of the 2D XY model across the BKT transition; a MALA correction after ODE sampling removes the discretization error that otherwise biases C_v.
- Generalized zero-shot from $L = 8$ to $L = 64$ without retraining, capturing the BKT transition at unseen lattice sizes.

Courtroom-inspired verification of financial misinformation [ICWSM 2026 workshop, DOI: 10.36190/2026.40]

- Addressed LLMs' tendency to accept mathematically coherent but contextually false financial claims: an LLM (DeepSeek-chat) generates opposing prosecution and defense arguments for each claim, and a frozen ModernBERT encoder with parallel BiGRUs and context-attention pooling weighs the evidence.
- 3rd place on the FMD Challenge 2026 leaderboard: accuracy 0.955, F1 0.9549 on the blind test set.

Indian Institute of Technology Roorkee Aug 2023 – Apr 2024
Bachelor's Thesis Researcher Advisor: Prof. Peayush Choubey

- Frustrated Josephson-junction arrays map onto long-range Ising models, which are costly to simulate with Monte Carlo; as a first step toward them, studied transfer from nearest-neighbour (NN) to next-nearest-neighbour (NNN) Ising models.
- Pre-trained deep networks on NN data and fine-tuned them on limited NNN data to predict phase transitions; compared fine-tuning configurations. Unlike earlier zero-shot NN→NNN transfer (Canabarro *et al.*, PRB 2019; Corte *et al.*, Comput. Mater. Sci. 2021), the network is adapted by fine-tuning.
- Earlier lab project (PHN300, 2023): classified Ising phases with feed-forward and convolutional networks trained on Monte Carlo configurations across lattice sizes.

McGill University, Montreal May 2023 – Aug 2023
MITACS Globalink Research Intern Advisor: Prof. John Kildea

- Monte Carlo (TOPAS) simulation of CR-39 track-etch detectors: reproduced published recoil-proton threshold energies across nine aluminium degrader thicknesses, and compared secondary-particle dose deposition from an isotropic monoenergetic neutron source with and without a water phantom.

Industry Experience

OLA Krutrim, Bangalore Jul 2024 – Dec 2025
Software Development Engineer I (AI Engineering & Research)

- Built production AI services: a RAG platform (Flask, MilvusDB retrieval, Celery/Redis), embedding and reranking models served on KServe/FastAPI at sub-200 ms latency, and NVIDIA DeepStream CCTV analytics over RTSP streams from 1000 service centres.
- Fine-tuned LLaMA-3-8B-Instruct with LoRA on 1000+ code Q&A pairs for repository understanding; SimPO preference optimization reduced hallucination, giving a 15% accuracy gain on paraphrased questions.

Selected Projects

- **Feature-based dynamic pricing** Feb 2024 – Mar 2024
  Course project, Reinforcement Learning (Prof. Manu Kumar Gupta): implemented ellipsoid and EXP4 contextual-bandit algorithms (Cohen *et al.*, 2020) and compared regret under changing distributions.
- **Monte Carlo simulation of magnetic skyrmions** Aug 2023 – Oct 2023
  Physics & Astronomy Club, IIT Roorkee: Metropolis sampling with simulated annealing to form topologically protected skyrmion configurations on a 2D square lattice.

Skills

Generative modeling	Score-based and diffusion models (DDPM, DDIM, manifold-aware SGMs), GANs, LoRA fine-tuning, SimPO, transformer encoders (ModernBERT), recurrent networks
Scientific computing	Monte Carlo methods (Metropolis, simulated annealing, MALA), TOPAS particle transport, statistical analysis of turbulence and spin-model observables
Software	Python, PyTorch, TensorFlow; Flask, FastAPI, Celery, Redis, MilvusDB, KServe, NVIDIA DeepStream

Selected Awards

- 3rd place (as independent researcher), Reference-Free Financial Misinformation Detection Challenge 2026
- GATE Data Science & AI: All India Rank 117 of 69,242 candidates 2026
- Keynote Speaker, NextGen Indian Leaders – DWDG: Addressed 200+ attendees on GenAI applications 2024
- Department Rank 2, B.Tech. Engineering Physics, IIT Roorkee 2024
- MITACS Globalink Research Internship, fully funded research at McGill University 2023
- ENCORE Award for All-Round Excellence, IIT Roorkee Alumni Awards 2022

Teaching & Service

- Student Mentor for 10+ first-year students, IIT Roorkee 2022–2024
- Undergraduate Teaching Assistant for Mathematics-I (70+ first-year students), IIT Roorkee 2021
- Deputy Cell Secretary, NSS IIT Roorkee: led 25 executives and 50 volunteers supporting 100+ underprivileged students; received the 'Best Cell Member' award 2021–2022

References

Prof. Rahul Pandit	National Science Chair, Department of Physics, IISc Bangalore	rahul@iisc.ac.in
Prof. Peayush Choubey	Assistant Professor, Department of Physics, IIT Roorkee	peayush@ph.iitr.ac.in
Prof. Manu Kumar Gupta	Asst. Professor, Mehta Family School of DS & ML, IIT Roorkee	manu.gupta@ms.iitr.ac.in